

Periodic ringsets in the Lipschitz quaternions: the complete six-letter classification

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Abstract

Let

$$\mathbf{L} = \mathbb{Z}[\mathbf{i}, \mathbf{j}, \mathbf{k}]$$

be the ring of Lipschitz quaternions, and let

$$U = \{\pm \mathbf{i}, \pm \mathbf{j}, \pm \mathbf{k}\}$$

be its six square roots of -1 . For a periodic word $w : \mathbb{Z} \rightarrow U$, we study

$$S_w = \{a + w(a) : a \in \mathbb{Z}\} \subseteq \mathbf{L}.$$

A subset of $\mathbf{L}$ is a ringset if the rational quaternion polynomials integer-valued on it are closed under multiplication. We give a complete criterion for S_w to be a ringset for every prescribed period, including even periods and words containing antipodal letters.

At ramified and nonsplit prime divisors of the period, every highest prime-power position class must contain at least two letters. At a split prime, the only obstruction occurs when two position fibres separated by a square root of -1 are singletons carrying antipodal letters. Necessity is proved by explicit null polynomials. The ramified prime 2 requires a Gaussian valuation calculation and a squared central fixed-divisor factor; the split obstruction uses rank-one idempotents over a matrix ring. These local results, periodic congruence orbits, and the Chinese remainder theorem give the global if-and-only-if classification. The theorem strictly generalizes the preceding squarefree-period and all-odd-period binary classifications.

1 Introduction

Let $\mathbb{D}$ be the rational Hamilton quaternion algebra and let

$$\mathbf{L} = \mathbb{Z}[\mathbf{i}, \mathbf{j}, \mathbf{k}], \quad \mathbf{i}^2 = \mathbf{j}^2 = -1, \quad \mathbf{ij} = \mathbf{k} = -\mathbf{ji}.$$

The variable in $\mathbb{D}[x]$ is central. Polynomials have coefficients on the left and are evaluated on the right:

$$\left(\sum_d c_d x^d \right) (\alpha) = \sum_d c_d \alpha^d.$$

For $S \subseteq \mathbf{L}$, put

$$\text{Int}(S, \mathbf{L}) = \{f \in \mathbb{D}[x] : f(S) \subseteq \mathbf{L}\}.$$

Following Werner [6], S is called a *ringset* if $\text{Int}(S, \mathbf{L})$ is a subring of $\mathbb{D}[x]$. This condition is nontrivial because the coefficient algebra is noncommutative.

Werner [6] classified finite ringsets in the Lipschitz and Hurwitz quaternion rings and supplied residue-null-ideal criteria for infinite sets. The first paper in the present series [1] classified periodic binary sets of odd squarefree period. The second [2] removed squarefreeness and classified all odd-period words with alphabet $\{\mathbf{i}, \mathbf{j}\}$. Two natural boundaries remained:

1. the ramified prime 2, which is unavoidable for even periods;
2. the other four Lipschitz square roots of -1 , particularly antipodal pairs at split primes.

This paper closes both boundaries. Let

$$U = \{\pm \mathbf{i}, \pm \mathbf{j}, \pm \mathbf{k}\},$$

let $m \geq 1$, and let $w : \mathbb{Z} \rightarrow U$ satisfy $w(a + m) = w(a)$. The displayed period is not required to be least. Define

$$S_w = \{a + w(a) : a \in \mathbb{Z}\}.$$

For a prime $p \mid m$, write $\nu_p = v_p(m)$. For $c \in \mathbb{Z}/p^{\nu_p}\mathbb{Z}$, define the position-fibre alphabet

$$A_p(c) = \{w(a) : a \equiv c \pmod{p^{\nu_p}}\} \subseteq U.$$

Our main result is the following.

Theorem 1.1. *Let $w : \mathbb{Z} \rightarrow U$ have period $m \geq 1$. Then S_w is a ringset in $\mathbf{L}$ if and only if all of the following hold:*

1. w is nonconstant;
2. for every prime $p \mid m$ such that either $p = 2$ or $p \equiv 3 \pmod{4}$, one has

$$|A_p(c)| \geq 2 \quad (c \in \mathbb{Z}/p^{\nu_p}\mathbb{Z});$$

3. for every prime $p \mid m$ with $p \equiv 1 \pmod{4}$, every $s \in \mathbb{Z}/p^{\nu_p}\mathbb{Z}$ satisfying $s^2 = -1$, every $c \in \mathbb{Z}/p^{\nu_p}\mathbb{Z}$, and every $u \in U$, it is not the case that

$$A_p(c - s) = \{u\} \quad \text{and} \quad A_p(c + s) = \{-u\}.$$

Condition 3 is independent of the choice between the two square roots: replacing s by $-s$ interchanges the fibres and replaces u by $-u$. All three conditions are independent of the choice of displayed period.

For the binary alphabet $\{\mathbf{i}, \mathbf{j}\}$, condition 3 is automatic, and Theorem 1.1 reduces on odd periods to [2, Theorem 1.1]. The new mechanisms are the ramified 2-adic obstruction and the

exact antipodal obstruction at split primes. The paired-fibre lemma, matrix representation, fixed-divisor background, and residue-null-ideal bridge are not claimed as new.

The rest of the paper proves Theorem 1.1. Sections 2 and 3 collect the global and paired-fibre tools. Section 4 isolates the split-prime geometry. Section 5 records a prime-power fixed-divisor estimate. Sections 6 and 7 construct the nonsplit and ramified obstruction polynomials. Section 8 assembles the classification.

2 Residue null ideals and periodic fibres

For a ring A and $T \subseteq A$, define

$$\mathcal{N}(T, A) = \{f \in A[x] : f(t) = 0 \text{ for every } t \in T\}.$$

This is always a left ideal. The set T is *core* if $\mathcal{N}(T, A)$ is a two-sided ideal. For $n \geq 2$, let $\bar{S}_n$ denote the image of $S \subseteq \mathbf{L}$ in $\mathbf{L}/n\mathbf{L}$.

We use the following consequence of Werner [6, Lemmas 5.3 and 5.4].

Lemma 2.1 (Residue criterion). *Let $S \subseteq \mathbf{L}$.*

1. *If $\bar{S}_n$ is core in $\mathbf{L}/n\mathbf{L}$ for every $n \geq 2$, then S is a ringset.*
2. *Conversely, if for some n there are $g \in \mathcal{N}(\bar{S}_n, \mathbf{L}/n\mathbf{L})$ and $r \in \mathbf{L}/n\mathbf{L}$ such that the coefficientwise right multiple gr is not null on $\bar{S}_n$, then S is not a ringset.*

Proof. Write a rational quaternion polynomial as $f = g/n$ with $g \in \mathbf{L}[x]$. By [6, Lemma 5.3], it is integer-valued on S exactly when the reduction of g is null on $\bar{S}_n$. Stability of every null ideal under coefficientwise right multiplication by constants gives multiplicative closure by [6, Lemma 5.4]. Conversely, a failed right multiple lifts to an integer-valued polynomial whose corresponding product is not integer-valued. $\square$

The alphabet U is intrinsic to the Lipschitz order.

Lemma 2.2. *The elements of $\mathbf{L}$ whose square is -1 are exactly $\pm \mathbf{i}, \pm \mathbf{j}, \pm \mathbf{k}$.*

Proof. If $q = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$ and $q^2 = -1$, its vector part is $2a(b\mathbf{i} + c\mathbf{j} + d\mathbf{k})$, so either $a = 0$ or the vector part vanishes. The latter is incompatible with $q^2 = -1$ for integral a . Thus $a = 0$ and $b^2 + c^2 + d^2 = 1$, whose integral solutions are the six stated units. $\square$

We next record the congruence orbit of a periodic word.

Lemma 2.3 (Periodic orbit lemma). *Let w have period m , let $q \geq 1$, and fix $a \in \mathbb{Z}/q\mathbb{Z}$. The positions modulo m attained by integers congruent to a modulo q are*

$$a + q\mathbb{Z} \pmod{m} = \{r \in \mathbb{Z}/m\mathbb{Z} : r \equiv a \pmod{\gcd(q, m)}\}.$$

Proof. The subgroup generated by q in $\mathbb{Z}/m\mathbb{Z}$ consists exactly of the multiples of $\gcd(q, m)$. Translate by a . $\square$

For $q = p^e$ and $\nu = v_p(m)$, a real residue fibre modulo p^e therefore sees one position class modulo $p^{\min(e, \nu)}$. In particular, a highest-level class modulo p^ν controls every finer real residue fibre, while a coarser fibre is a union of highest-level classes.

The same formula proves invariance under a nonminimal displayed period. If $d \mid m$ is the least period, then a class modulo $p^{v_p(m)}$ visits exactly a class modulo $p^{v_p(d)}$ when $p \mid d$, and visits the entire least period when $p \nmid d$. The letter sets in Theorem 1.1 therefore repeat, rather than change, when the displayed period is enlarged. At split primes, Hensel lifting transports the two square-root offsets compatibly.

3 More than one letter above every real residue

Every point $a + u$ with $u \in U$ has central minimal polynomial

$$\mu_a(x) = (x - a)^2 + 1.$$

The following paired-fibre lemma extends the calculation used in [1, 2].

Lemma 3.1 (Multi-letter paired fibres). *Let $n \geq 2$, and let*

$$T \subseteq \{a + u : a \in \mathbb{Z}/n\mathbb{Z}, u \in U\}.$$

If at least two distinct letters occur above every real residue, then T is core in $\mathbf{L}/n\mathbf{L}$.

Proof. Take $f \in \mathcal{N}(T, \mathbf{L}/n\mathbf{L})$ and fix a real residue a . Divide f by the central monic polynomial μ_a . The remainder is

$$r(x) = \gamma_1 x + \gamma_0.$$

If $a + u$ and $a + v$ are two points of T with $u \neq v$, subtraction of their zero evaluations gives

$$\gamma_1(u - v) = 0.$$

If $v \neq -u$, then $\text{Nm}(u - v) = 2$, so right multiplication by $\overline{u - v}$ yields $2\gamma_1 = 0$. If $v = -u$, then $u - v = 2u$, and right multiplication by the unit $-u$ gives the same conclusion.

For a basis unit $t \in \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ and $\alpha = a + u$, direct right evaluation gives

$$(rt)(\alpha) - r(\alpha)t = \gamma_1(t\alpha - \alpha t).$$

Every displayed commutator lies in $2\mathbf{L}/n\mathbf{L}$, so the right-hand side vanishes. Centrality of μ_a passes this conclusion from the remainder to f . Thus the null ideal is stable under coefficientwise right multiplication by the quaternion basis and hence by every constant. Since it is a left ideal and x is central, it is then stable under right multiplication by every polynomial. $\square$

Lemma 3.1 immediately supplies all ramified and nonsplit sufficiency directions in Theorem 1.1. The remaining sufficiency issue is the split case, where a single letter above a real residue can already suffice.

4 Split primes and antipodal fibres

Let $p \equiv 1 \pmod{4}$, let $e \geq 1$, put

$$R = \mathbb{Z}/p^e\mathbb{Z}, \quad A = \mathbf{L}/p^e\mathbf{L},$$

and choose $s \in R$ with $s^2 = -1$. The standard representation

$$\mathbf{i} \mapsto \begin{pmatrix} s & 0 \\ 0 & -s \end{pmatrix}, \quad \mathbf{j} \mapsto \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

identifies A with $M_2(R)$. For $u \in U$, define

$$e_u^+ = \frac{1 + u/s}{2}, \quad e_u^- = \frac{1 - u/s}{2}.$$

They are complementary rank-one idempotents, and

$$(a + u)^d = (a + s)^d e_u^+ + (a - s)^d e_u^-.$$

Generators for their image lines may be chosen as follows.

u	$\text{im}(e_u^+)$	$\text{im}(e_u^-)$
$\mathbf{i}$	$(1, 0)^T$	$(0, 1)^T$
$-\mathbf{i}$	$(0, 1)^T$	$(1, 0)^T$
$\mathbf{j}$	$(1, s)^T$	$(1, -s)^T$
$-\mathbf{j}$	$(1, -s)^T$	$(1, s)^T$
$\mathbf{k}$	$(1, 1)^T$	$(1, -1)^T$
$-\mathbf{k}$	$(1, -1)^T$	$(1, 1)^T$

The determinants obtained from distinct same-sign lines are units. A plus line for u and a minus line for v fail to be complementary exactly when $v = -u$. Indeed, all nonzero determinants are among

$$\pm 1, \quad \pm s, \quad \pm 2, \quad \pm 2s, \quad \pm(1 + s), \quad \pm(1 - s),$$

and these are units because p is odd and $(1 + s)(1 - s) = 2$.

Lemma 4.1 (Split local core criterion). *Let*

$$T \subseteq \{a + u : a \in R, u \in U\}$$

have a nonempty fibre above every $a \in R$. Suppose that, for every $z \in R$, the two fibres above $z - s$ and $z + s$ are not respectively the singleton pair $\{u\}$ and $\{-u\}$ for any $u \in U$. Then T is core in A .

Proof. Take a null polynomial $f(x) = \sum_d c_d x^d$ and introduce the scalar-variable matrix polynomial

$$F(z) = \sum_d c_d z^d.$$

At fixed z , every letter u occurring above $z - s$ gives

$$F(z)e_u^+ = 0,$$

and every letter v occurring above $z + s$ gives

$$F(z)e_v^- = 0.$$

If either fibre contains two letters, the corresponding two same-sign image lines are complementary. If both are singletons, the hypothesis and the table show that the plus and minus lines are complementary. Thus $F(z) = 0$ for every $z \in R$.

For any $B \in A$, the scalar polynomial associated with the coefficientwise right multiple fB is $F(z)B = 0$. The same idempotent decomposition shows that fB vanishes on T . Hence the null ideal is stable under all constant right multiples and therefore is two-sided. $\square$

The excluded singleton-antipode configuration is also a genuine periodic obstruction. We first record the fixed-divisor estimate used to detect it.

5 A prime-power fixed-divisor estimate

Lemma 5.1. *Let p be a prime, let $\nu \geq 1$, and put*

$$M = p^\nu, \quad V = v_p((M-1)!), \quad P_c(t) = \prod_{r=1}^{M-1} (t - (c+r)).$$

Then

$$v_p(P_c(t)) = \begin{cases} V, & t \equiv c \pmod{M}, \\ \geq V+1, & t \not\equiv c \pmod{M}. \end{cases}$$

Proof. The assertion is translation-invariant, so write $d = t - c$. If $M \mid d$, then $v_p(d-r) = v_p(r)$ for $1 \leq r < M$, giving V .

Otherwise choose $h \in \{1, \dots, M-1\}$ with $d \equiv h \pmod{M}$. The factor $d-h$ contributes at least ν , while comparison of the two remaining factorial segments gives

$$v_p(P_c(t)) \geq \nu + v_p((h-1)!) + v_p((M-1-h)!).$$

Because $\binom{M-1}{h}$ is nonzero modulo p ,

$$v_p((h-1)!) + v_p((M-1-h)!) = V - v_p(h).$$

Since $v_p(h) \leq \nu - 1$, the result follows. $\square$

Set $K = V + 1$. Notice that $K \geq \nu$: this is immediate for $\nu = 1$, and for $\nu \geq 2$ one has

$$V \geq \left\lfloor \frac{p^\nu - 1}{p} \right\rfloor = p^{\nu-1} - 1 \geq \nu - 1.$$

Lemma 5.2 (Split antipodal obstruction). *Let $p \equiv 1 \pmod{4}$, let $\nu = v_p(m) \geq 1$, and choose $s \in \mathbb{Z}/p^\nu\mathbb{Z}$ with $s^2 = -1$. If for some c and $u \in U$*

$$A_p(c - s) = \{u\}, \quad A_p(c + s) = \{-u\},$$

then S_w is not a ringset.

Proof. Let $M = p^\nu$, let V, K be as above, and Hensel-lift s to a square root of -1 modulo p^K . In

$$\mathbf{L}/p^K\mathbf{L} \simeq M_2(\mathbb{Z}/p^K\mathbb{Z}),$$

put $E = e_u^+$ and $A = 1 - E = e_u^-$. Choose a matrix B such that

$$ABE \not\equiv 0 \pmod{p}.$$

For example, in a basis where $E = E_{11}$ one may take $A = E_{22}$ and $B = E_{21}$.

Consider the matrix-coefficient polynomial

$$F(t) = P_c(t)A.$$

At a point $a + v$, its idempotent decomposition is

$$F(a + v) = P_c(a + s)Ae_v^+ + P_c(a - s)Ae_v^-.$$

By Lemma 5.1, a term can survive modulo p^K only if its scalar parameter is congruent to c modulo M . The plus case then has $a \equiv c - s$ and $v = u$, so $Ae_u^+ = AE = 0$. The minus case has $a \equiv c + s$ and $v = -u$, so $e_{-u}^- = e_u^+ = E$ and is also killed. Thus F is null on the periodic image modulo p^K .

After coefficientwise right multiplication by B , evaluation at the appearing point $c - s + u$ has plus component

$$P_c(c)ABE.$$

Its p -adic valuation is exactly $V = K - 1$, because ABE is nonzero modulo p . Hence the right multiple is not null modulo p^K . Lemma 2.1 proves that S_w is not a ringset. $\square$

6 Nonsplit prime-power obstructions

Let $p \equiv 3 \pmod{4}$, and suppose that $A_p(c) = \{q\}$ for some $q \in U$. Put $M = p^\nu$, where $\nu = v_p(m)$, and retain V, K from Section 5. Define central polynomials

$$\mu_{c+r}(x) = (x - (c + r))^2 + 1, \quad Q_c(x) = \prod_{r=1}^{M-1} \mu_{c+r}(x),$$

and

$$h_c(x) = (x - (c + q))Q_c(x).$$

Lemma 6.1 (Higher nonsplit obstruction). *The polynomial h_c is null on the periodic image modulo p^K , but a suitable coefficientwise right multiple is not null. Consequently S_w is not a ringset.*

Proof. At $y = a + u$, with $d = a - c$, one has

$$\mu_{c+r}(y) = (d - r)(d - r + 2u).$$

The second factor is a unit modulo p , because its norm is $(d - r)^2 + 4$, which cannot vanish when -1 is nonsquare modulo p . Hence

For a Lipschitz quaternion z , write $v_p(z)$ for the minimum of the p -adic valuations of its four integer coordinates, equivalently for the largest s such that $z \in p^s \mathbf{L}$. Then

$$v_p(Q_c(y)) = v_p(P_c(a)).$$

If $M \nmid d$, Lemma 5.1 makes $Q_c(y)$ zero modulo p^K . If $M \mid d$, the periodic orbit lemma says that the only occurring letter is q ; the leading factor then evaluates to the real integer d , whose valuation is at least ν . Since $V + \nu \geq K$, $h_c(y) = 0$ in every case.

Choose a basis unit v that does not commute with q . Since Q_c is central,

$$h_c v = ((x - (c + q))v)Q_c.$$

At the appearing point $c + q$, the first factor evaluates to

$$v(c + q) - (c + q)v = vq - qv,$$

which is twice a quaternion unit and hence a p -adic unit. The second factor has valuation exactly $V = K - 1$. Thus $h_c v$ is not null modulo p^K . Apply Lemma 2.1. $\square$

A constant word may be displayed with period 3. Taking $p = 3$ and $\nu = 1$ gives $A_3(c) = \{q\}$ for every position class c , so Lemma 6.1 supplies a constant-word obstruction modulo 3. Therefore nonconstancy in Theorem 1.1 is necessary even when the originally displayed period has no prime divisor requiring a separate condition.

7 The ramified prime 2

The proof at 2 differs because -1 has ramified rather than split or nonsplit behaviour. Let $\nu = v_2(m) \geq 1$, and suppose that one position class modulo 2^ν contains only a letter $q \in U$. Choose

$$N \geq \max\{\nu, 3\}, \quad M = 2^N,$$

and refine the monochromatic class to a class $c \pmod{M}$. Every occurring point whose real part is congruent to c modulo M still has letter q . Put

$$Q_c(x) = \prod_{r=1}^{M-1} ((x - (c + r))^2 + 1),$$

$$h_c(x) = (x - (c + q))Q_c(x)^2.$$

The square on the central factor is the essential ramified correction.

For $u \in U$, work in the Gaussian subring $\mathbb{Z}[u]$ and write $\pi = 1 + u$. Since 2 is associated with π^2 , divisibility of all four Lipschitz coordinates by 2^k is equivalent, for elements of $\mathbb{Z}[u]$, to π -adic valuation at least $2k$.

Let

$$P(d) = \prod_{r=1}^{M-1} (d - r), \quad V = v_2((M-1)!).$$

At $y = a + u$, with $d = a - c$, direct calculation gives

$$Q_c(y) = P(d)P(d + 2u).$$

Define

$$\tau(n) = v_2(n^2 + 4).$$

Then

$$v_\pi(Q_c(y)) = 2v_2(P(d)) + \sum_{r=1}^{M-1} \tau(d - r).$$

The value of τ depends only on $n \pmod{4}$:

$$\tau(n) = \begin{cases} 0, & n \text{ odd}, \\ 3, & n \equiv 2 \pmod{4}, \\ 2, & n \equiv 0 \pmod{4}. \end{cases}$$

If $M \mid d$, the baseline valuation is

$$L_0 = 2V + \frac{5M}{4} - 2.$$

Indeed, among $1, \dots, M-1$ there are $M/2$ odd residues, $M/4$ residues congruent to 2 modulo 4, and $M/4 - 1$ nonzero residues divisible by 4.

Lemma 7.1 (Ramified valuation gap). *If $M \nmid d$, then*

$$v_\pi(Q_c(a + u)) \geq L_0 + 2.$$

Proof. Choose $s \in \{1, \dots, M-1\}$ with $d \equiv s \pmod{M}$. The consecutive integer product satisfies

$$v_2(P(d)) \geq V - v_2(s) + N.$$

Compared with the baseline residue set, the τ -sum gains 2 if s is odd, loses 1 if $s \equiv 2 \pmod{4}$, and is unchanged if $s \equiv 0 \pmod{4}$. Consequently the total gain is at least

$$2N + 2, \quad 2N - 3, \quad 2,$$

respectively. The last estimate uses $v_2(s) \leq N - 1$, and the second is at least 3 because $N \geq 3$. $\square$

Lemma 7.2 (Ramified monochromatic obstruction). *If a highest 2-power position class is monochromatic, then S_w is not a ringset.*

Proof. Set $K = L_0 + 2$. Since $K \geq \nu$, the periodic orbit in the quotient modulo 2^K sees the original class modulo 2^ν .

If $M \nmid d$, Lemma 7.1 gives

$$v_\pi(Q_c(y)^2) \geq 2(L_0 + 2),$$

so every Lipschitz coordinate of $Q_c(y)^2$ is divisible by 2^K . If $M \mid d$, the occurring letter is q and the leading factor of $h_c(y)$ is the real integer d , divisible by 2^N . The coordinate valuation of $Q_c(y)^2$ is L_0 , so again $h_c(y) = 0 \pmod{2^K}$. Thus h_c is null on the entire periodic image.

Choose a basis unit v transverse to q . Since Q_c^2 is central,

$$h_c v = ((x - (c + q))v)Q_c^2.$$

At $c + q$, the leading factor is the commutator $vq - qv$, whose coordinate 2-valuation is exactly 1. The coordinate valuation of $Q_c(c + q)^2$ is exactly L_0 . Therefore

$$v_2((h_c v)(c + q)) = L_0 + 1 < K,$$

and the right multiple is not null modulo 2^K . Lemma 2.1 completes the proof. $\square$

Together with Lemma 3.1, Lemma 7.2 gives the exact local condition at 2: all residue images modulo powers of 2 are core if and only if every highest 2-power position class contains at least two letters.

8 Proof of the classification

Proof of Theorem 1.1. We now prove Theorem 1.1.

Sufficiency.

Assume conditions 1–3. By Lemma 2.1 and the Chinese remainder theorem, it is enough to prove that the periodic image is core modulo every prime power p^e .

Suppose first that $p = 2$. If $p \nmid m$, every real residue fibre traverses the entire word by Lemma 2.3 and therefore contains at least two letters by nonconstancy. If $p \mid m$, condition 2 makes every highest position class contain at least two letters, and hence every coarser real fibre does also. Lemma 3.1 applies.

The same argument applies when $p \equiv 3 \pmod{4}$: use nonconstancy when $p \nmid m$, and condition 2 when $p \mid m$. Again Lemma 3.1 gives core images.

Now let $p \equiv 1 \pmod{4}$. If $p \nmid m$, both fibres used at each scalar parameter in Lemma 4.1 traverse the full nonconstant word, so each contains at least two letters. If $p \mid m$ and $e \geq \nu_p$, their letter sets are the highest position fibres in condition 3, after projecting the chosen square root modulo p^e to modulo p^{ν_p} . Lemma 4.1 applies directly.

If $e < \nu_p$ and the two coarse fibres were singleton antipodes, every highest-level class refining the first would carry the same singleton, and every class refining the second would carry its antipode. Hensel-lift the root modulo p^e to the unique root modulo p^{ν_p} and lift the scalar centre arbitrarily. This would violate condition 3. Hence Lemma 4.1 applies for every e .

For an arbitrary $n = \prod_r p_r^{e_r}$, the Chinese remainder theorem gives

$$\mathbf{L}/n\mathbf{L} \simeq \prod_r \mathbf{L}/p_r^{e_r}\mathbf{L}.$$

Polynomial coefficients and evaluation are componentwise, so the null ideal of the image is the product of the null ideals of its projections. Every prime-power projection is core, and Lemma 2.1 proves that S_w is a ringset.

Necessity.

If w is constant, display it with period 3 and take $p = 3$, $\nu = 1$. Every position fibre is then the singleton $\{q\}$, so Lemma 6.1 shows that S_w is not a ringset.

If condition 2 fails for a prime $p \equiv 3 \pmod{4}$, Lemma 6.1 gives an explicit null polynomial and a failed right multiple. If it fails for $p = 2$, Lemma 7.2 gives the corresponding ramified witness. In both cases Lemma 2.1 shows that S_w is not a ringset.

Finally, if condition 3 fails at a split prime, Lemma 5.2 gives an explicit matrix-coefficient null polynomial whose coefficientwise right multiple is not null. Thus S_w is again not a ringset. All three conditions are necessary, completing the proof. $\square$

9 Consequences and examples

Corollary 9.1 (Binary words). *Let $w : \mathbb{Z} \rightarrow \{\mathbf{i}, \mathbf{j}\}$ have arbitrary period m . Then S_w is a ringset if and only if w is nonconstant and, for every $p \mid m$ with $p = 2$ or $p \equiv 3 \pmod{4}$, each position class modulo $p^{\nu_p(m)}$ is bichromatic.*

Proof. The binary alphabet has no antipodal pair, so condition 3 of Theorem 1.1 is automatic. $\square$

Corollary 9.1 strictly extends [2], which assumes that m is odd. In particular, the condition at 2 is not a formal copy of the odd-prime condition: its necessity comes from the ramified Gaussian calculation in Section 7.

Corollary 9.2 (Antipode-free alphabets). *Let $\Omega \subseteq U$ contain at least two letters and no pair $u, -u$. For a periodic word $w : \mathbb{Z} \rightarrow \Omega$, the set S_w is a ringset if and only if conditions 1 and 2 of Theorem 1.1 hold.*

The split condition disappears because no forbidden antipodal singleton pair can occur. This includes the natural three-letter alphabet $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$.

The full alphabet exhibits genuinely new split failures. If $p \equiv 1 \pmod{4}$ divides the period and two square-root-separated position classes are constantly u and $-u$, respectively, then all ramified and nonsplit colour conditions may hold while the set still fails to be a ringset. Lemma 5.2 detects exactly this configuration.

10 Computational calibration

The proof is symbolic and does not depend on finite enumeration. Computation was used during discovery to locate and attack the local boundaries.

- All 81 three-state fibre patterns modulo 4 and all 256 binary words of period 8 were checked against the evaluation kernel in degrees strictly below $2q$ for the tested modulus q .
- The 16 period-8 patterns surviving modulo 16 all failed modulo 64, disproving the provisional half-period-antipodal sufficiency guess.
- The ramified valuation formula was independently checked for $N = 3, \dots, 10$; this exposed that the smallest off-class gain is 2, not 3, and the proof above uses the corrected value.
- The split eigenline table was checked for all 36 ordered letter pairs; exactly the six antipodal pairs fail complementarity.
- The split obstruction was evaluated for $p = 5, 13, 17$ and $\nu = 1, 2$ in finite quotients.

These calculations are calibration and destructive testing only. They do not establish the all-period theorem, whose general content is proved in Sections 3–8. The first two bullets share the same truncated-null-module solver and therefore are not independent implementations.

11 Prior work and novelty boundary

Frisch [4] established a general connection between ringsets and null-ideal sets. Werner [5] classified null ideals of subsets of matrix rings over fields, while Rissner and Werner [7] studied their enumeration. Werner [6] classified finite Lipschitz and Hurwitz quaternion ringsets and proved the residue and right-multiplication criteria used here. Peruginelli [3] developed prime-power fixed-divisor ideals in the commutative setting. None of these general tools is claimed as new.

The preceding squarefree-period paper [1] supplies the initial periodic framework, and [2] proves the complete all-odd-period binary classification. The present theorem strictly contains both published results. Its candidate new content is limited to:

1. the explicit ramified 2-adic monochromatic-fibre obstruction;
2. the exact split-prime antipodal singleton obstruction;
3. their assembly into the complete six-letter, arbitrary-period classification.

The paired-fibre calculation, split matrix model, idempotent vocabulary, consecutive-product fixed-divisor estimates, and Chinese-remainder assembly are treated as prior tools or standard ingredients.

As of the bounded search completed on 24 August 2026, no public source was found that directly states or implies the complete criterion of Theorem 1.1. This is a qualified `NO_DIRECT_COVERAGE_FOUND` statement, not a certificate of absolute global priority. The manuscript has not received external expert review or formal peer review.

Declaration of generative AI and AI-assisted technologies

During the research and preparation of this work, the author used OpenAI Codex to assist with literature searches, symbolic exploration, exploratory code review, proof checking, and language drafting. The author reviewed and edited all generated output and takes full responsibility for the content of this manuscript.

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